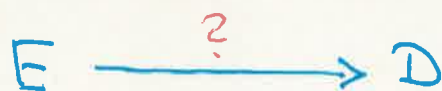


Situation:



wanted:

$$CRR = \frac{P(Y^{a=1} = 1)}{P(Y^{a=0} = 1)} = \frac{P(D^{e=1} = 1)}{P(D^{e=0} = 1)}$$

↑

Notation of Jewell (2004)

Problem: CRR cannot be computed because of (missing) counterfactuals ~~and~~ and sample data

Available:

$$ARR = \frac{P(Y=1 | A=1)}{P(Y=1 | A=0)}$$

"Take-home message 1":

Ideal randomized experiment

→ Exchangeability → $ARR = CRR$
("Association is causation")

Example of Zeus's family:

$$ARR = 1.26 \neq 1 = CRR$$

⇒ Groups are not exchangeable.

Standardization:

$$\text{CRR} = \frac{P(Y^{a=1}=1)}{P(Y^{a=0}=1)} = \frac{\sum_l P(Y=1 | A=1, L=l) \cdot P(L=l)}{\sum_l P(Y=1 | A=0, L=l) \cdot P(L=l)}$$

$\uparrow$
 $Y^a \perp\!\!\!\perp A | L$

$$= \frac{\frac{1}{4} \cdot \frac{8}{20} + \frac{6}{9} \cdot \frac{12}{20}}{\frac{1}{4} \cdot \frac{8}{20} + \frac{2}{3} \cdot \frac{12}{20}} = 1$$

Inverse probability weighting (IPW)

$$\text{CRR} = \frac{P(Y^{a=1}=1)}{P(Y^{a=0}=1)} = \frac{10/20}{10/20} = 1$$

$\nwarrow$ Right panel
 $\nearrow$ Left panel

"Take-home message 2"

Ideal conditionally randomized experiment

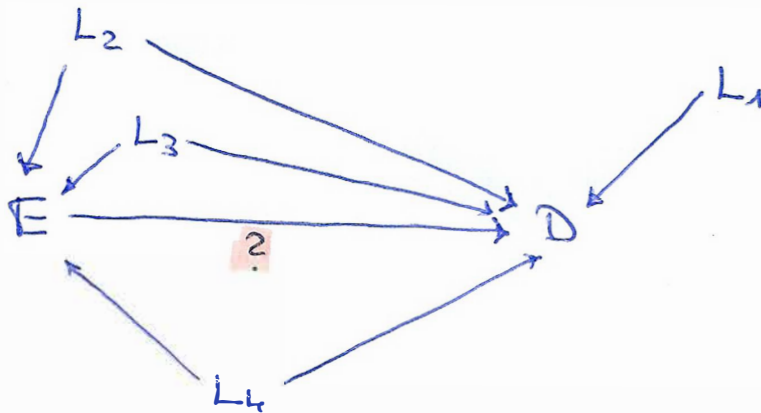
→ Conditional exchangeability

→ Use of standardization or IPW → CRR

Objective: To study whether E (certain drug) increases risk for D (certain cancer).

Assumptions: D has known risk factors L_1 (Pollution), L_2 (Age), L_3 (Gender), and L_4 (Gene mutation)

L_2, L_3, L_4 are related to E



That is, $D(L_1 | E) = D(L_1 | \bar{E})$

(D: distribution) $D(L_2 | E) \neq D(L_2 | \bar{E})$

$D(L_3 | E) \neq D(L_3 | \bar{E})$

$D(L_4 | E) \neq D(L_4 | \bar{E})$

Causal inference is possible, if information on L_2, L_3 , and L_4 is available. Information on L_1 is not necessary to estimate ERR.

CRR can be computed/estimated using

(i) Standardization

→ Logistic regression model including E, L_2, L_3 , and L_4 as independent variables.

(ii) IPW using weights $\frac{1}{P(E | L_2, L_3, L_4)}$

But: Can we be sure, there are no other risk factors of D that are also related to E?